2.33 Reduction Formulae for Integration

Worked Solutions

1)

Let $I_n = \int x^n e^x \, dx$. By integration by parts, take $u = x^n$ and $dv = e^x \, dx$. Then $du = nx^{n-1} \, dx$ and $v = e^x$, so

$$I_n = x^n e^x - n \int x^{n-1} e^x \, dx = x^n e^x - n I_{n-1},$$

as required.

2a)

$$I_0 = \int e^x \, dx = e^x + c.$$

Then

$$\begin{aligned} I_1 &= xe^x - I_0 = xe^x - e^x + c = e^x(x-1) + c, \\ I_2 &= x^2 e^x - 2I_1 = e^x(x^2 - 2x + 2) + c. \end{aligned}$$

2b)

$$\begin{aligned} I_0 &= e^x + c, \\ I_1 &= xe^x - I_0 = e^x(x-1) + c, \\ I_2 &= x^2 e^x - 2I_1 = e^x(x^2 - 2x + 2) + c, \\ I_3 &= x^3 e^x - 3I_2 = e^x(x^3 - 3x^2 + 6x - 6) + c. \end{aligned}$$

2c)

$$\begin{aligned} I_0 &= e^x + c, \\ I_1 &= xe^x - I_0 = e^x(x-1) + c, \\ I_2 &= x^2 e^x - 2I_1 = e^x(x^2 - 2x + 2) + c, \\ I_3 &= x^3 e^x - 3I_2 = e^x(x^3 - 3x^2 + 6x - 6) + c, \\ I_4 &= x^4 e^x - 4I_3 = e^x(x^4 - 4x^3 + 12x^2 - 24x + 24) + c. \end{aligned}$$

3)

Let $I_n = \int x^n e^{2x} \, dx$. By integration by parts, take $u = x^n$ and $dv = e^{2x} \, dx$. Then $du = nx^{n-1} \, dx$ and $v = \frac{1}{2}e^{2x}$, so

$$I_n = \frac{1}{2}x^n e^{2x} - \frac{n}{2} \int x^{n-1} e^{2x} \, dx = \frac{1}{2}x^n e^{2x} - \frac{n}{2} I_{n-1},$$

as required.

4a)

$$I_0 = \int e^{2x}\,dx = \frac{1}{2}e^{2x} + c.$$

Using the reduction formula,

$$\begin{aligned}
I_1 &= \frac{1}{2}xe^{2x} - \frac{1}{2}I_0 = \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + c,\\
I_2 &= \frac{1}{2}x^2e^{2x} - I_1 = \frac{1}{2}x^2e^{2x} - \frac{1}{2}xe^{2x} + \frac{1}{4}e^{2x} + c,\\
I_3 &= \frac{1}{2}x^3e^{2x} - \frac{3}{2}I_2\\
&= \frac{1}{2}x^3e^{2x} - \frac{3}{4}x^2e^{2x} + \frac{3}{4}xe^{2x} - \frac{3}{8}e^{2x} + c\\
&= \frac{1}{8}e^{2x}(4x^3 - 6x^2 + 6x - 3) + c.
\end{aligned}$$

4b)

$$\begin{aligned}
I_0 &= \frac{1}{2}e^{2x} + c,\\
I_1 &= \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + c,\\
I_2 &= \frac{1}{2}x^2e^{2x} - \frac{1}{2}xe^{2x} + \frac{1}{4}e^{2x} + c,\\
I_3 &= \frac{1}{2}x^3e^{2x} - \frac{3}{4}x^2e^{2x} + \frac{3}{4}xe^{2x} - \frac{3}{8}e^{2x} + c,\\
I_4 &= \frac{1}{2}x^4e^{2x} - 2I_3\\
&= \frac{1}{2}x^4e^{2x} - x^3e^{2x} + \frac{3}{2}x^2e^{2x} - \frac{3}{2}xe^{2x} + \frac{3}{4}e^{2x} + c\\
&= \frac{1}{4}e^{2x}(2x^4 - 4x^3 + 6x^2 - 6x + 3) + c.
\end{aligned}$$

5)

Let $I_n = \int x(\ln x)^n\,dx$. By integration by parts, take $u = (\ln x)^n$ and $dv = x\,dx$. Then $du = \frac{n}{x}(\ln x)^{n-1}\,dx$ and $v = \frac{1}{2}x^2$. Therefore

$$\begin{aligned}
I_n &= \frac{x^2}{2}(\ln x)^n - \frac{n}{2}\int x(\ln x)^{n-1}\,dx\\
&= \frac{x^2}{2}(\ln x)^n - \frac{n}{2}I_{n-1},
\end{aligned}$$

as required.

6a)

$$I_0 = \int x\,dx = \frac{x^2}{2} + c.$$

Hence

$$I_1 = \frac{x^2}{2}\ln x - \frac{1}{2}I_0 = \frac{x^2}{2}\ln x - \frac{x^2}{4} + c,$$

$$
\begin{aligned}
I_2 &= \frac{x^2}{2}(\ln x)^2 - I_1 \\
&= \frac{x^2}{2}(\ln x)^2 - \frac{x^2}{2}\ln x + \frac{x^2}{4} + c \\
&= \frac{x^2}{4}\Big(2(\ln x)^2 - 2\ln x + 1\Big) + c.
\end{aligned}
$$

6b)

$$
\begin{aligned}
I_0 &= \frac{x^2}{2} + c, \\
I_1 &= \frac{x^2}{2}\ln x - \frac{x^2}{4} + c, \\
I_2 &= \frac{x^2}{2}(\ln x)^2 - \frac{x^2}{2}\ln x + \frac{x^2}{4} + c, \\
I_3 &= \frac{x^2}{2}(\ln x)^3 - \frac{3}{2}I_2 \\
&= \frac{x^2}{2}(\ln x)^3 - \frac{3x^2}{4}(\ln x)^2 + \frac{3x^2}{4}\ln x - \frac{3x^2}{8} + c \\
&= \frac{x^2}{8}\Big(4(\ln x)^3 - 6(\ln x)^2 + 6\ln x - 3\Big) + c.
\end{aligned}
$$

7)

Let $I_n = \int \sin^n x\,dx = \int \sin x \sin^{n-1} x\,dx$. Use integration by parts with $u = \sin^{n-1} x$ and $dv = \sin x\,dx$. Then $du = (n-1)\cos x \sin^{n-2} x\,dx$ and $v = -\cos x$, so

$$
\begin{aligned}
I_n &= -\sin^{n-1} x\cos x + (n-1)\int \cos^2 x \sin^{n-2} x\,dx \\
&= -\sin^{n-1} x\cos x + (n-1)\int (1-\sin^2 x)\sin^{n-2} x\,dx \\
&= -\sin^{n-1} x\cos x + (n-1)I_{n-2} - (n-1)I_n.
\end{aligned}
$$

Thus $nI_n = -\sin^{n-1} x\cos x + (n-1)I_{n-2}$, and

$$
I_n = -\frac{\sin^{n-1} x\cos x}{n} + \frac{n-1}{n}I_{n-2},
$$

as required.

8a)

Using $I_0 = x + c$ and the reduction formula,

$$
\begin{aligned}
I_2 &= -\frac{\sin x\cos x}{2} + \frac{1}{2}I_0 = -\frac{\sin x\cos x}{2} + \frac{x}{2} + c, \\
I_4 &= -\frac{\sin^3 x\cos x}{4} + \frac{3}{4}I_2 \\
&= -\frac{\sin^3 x\cos x}{4} - \frac{3\sin x\cos x}{8} + \frac{3x}{8} + c.
\end{aligned}
$$

8b)

Using $I_1 = -\cos x + c$ and the reduction formula,

$$\begin{aligned} I_3 &= -\frac{\sin^2 x \cos x}{3} + \frac{2}{3} I_1 = -\frac{\sin^2 x \cos x}{3} - \frac{2}{3}\cos x + c, \\ I_5 &= -\frac{\sin^4 x \cos x}{5} + \frac{4}{5} I_3 \\ &= -\frac{\sin^4 x \cos x}{5} - \frac{4\sin^2 x \cos x}{15} - \frac{8\cos x}{15} + c. \end{aligned}$$

9)

Let $I_n = \int \cos^n x\, dx = \int \cos x \cos^{n-1} x\, dx$. Use integration by parts with $u = \cos^{n-1} x$ and $dv = \cos x\, dx$. Then $du = -(n-1)\sin x \cos^{n-2} x\, dx$ and $v = \sin x$, so

$$\begin{aligned} I_n &= \sin x \cos^{n-1} x + (n-1)\int \sin^2 x \cos^{n-2} x\, dx \\ &= \sin x \cos^{n-1} x + (n-1)\int (1 - \cos^2 x)\cos^{n-2} x\, dx \\ &= \sin x \cos^{n-1} x + (n-1) I_{n-2} - (n-1) I_n. \end{aligned}$$

Thus $nI_n = \sin x \cos^{n-1} x + (n-1)I_{n-2}$, and

$$I_n = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2},$$

as required.

10a)

Using $I_0 = x + c$ and the reduction formula,

$$\begin{aligned} I_2 &= \frac{\sin x \cos x}{2} + \frac{1}{2} I_0 = \frac{\sin x \cos x}{2} + \frac{x}{2} + c, \\ I_4 &= \frac{\sin x \cos^3 x}{4} + \frac{3}{4} I_2 \\ &= \frac{\sin x \cos^3 x}{4} + \frac{3\sin x \cos x}{8} + \frac{3x}{8} + c. \end{aligned}$$

10b)

Using $I_1 = \sin x + c$ and the reduction formula,

$$\begin{aligned} I_3 &= \frac{\sin x \cos^2 x}{3} + \frac{2}{3} I_1 = \frac{\sin x \cos^2 x}{3} + \frac{2}{3}\sin x + c, \\ I_5 &= \frac{\sin x \cos^4 x}{5} + \frac{4}{5} I_3 \\ &= \frac{\sin x \cos^4 x}{5} + \frac{4\sin x \cos^2 x}{15} + \frac{8\sin x}{15} + c. \end{aligned}$$

11a)

Let $I_n = \int \sinh^n x\, dx = \int \sinh x \sinh^{n-1} x\, dx$. Use integration by parts with $u = \sinh^{n-1} x$ and $dv = \sinh x\, dx$. Then $du = (n-1)\cosh x \sinh^{n-2} x\, dx$ and $v = \cosh x$, so

$$I_n = \cosh x \sinh^{n-1} x - (n-1)\int \cosh^2 x \sinh^{n-2} x\, dx$$

$$= \cosh x \sinh^{n-1} x - (n-1)I_{n-2} - (n-1)I_n.$$

Thus $nI_n = \cosh x \sinh^{n-1} x - (n-1)I_{n-2}$, giving

$$I_n = \frac{\sinh^{n-1} x \cosh x}{n} - \frac{n-1}{n} I_{n-2}.$$

11b)

$$I_0 = \int 1\, dx = x + c, \qquad I_2 = \frac{\sinh x \cosh x}{2} - \frac{1}{2} I_0.$$

Therefore

$$\begin{aligned} I_4 &= \frac{\sinh^3 x \cosh x}{4} - \frac{3}{4} I_2 \\ &= \frac{\sinh^3 x \cosh x}{4} - \frac{3 \sinh x \cosh x}{8} + \frac{3x}{8} + c. \end{aligned}$$

12a)

Let $I_n = \int \cosh^n x\, dx = \int \cosh x \cosh^{n-1} x\, dx$. Use integration by parts with $u = \cosh^{n-1} x$ and $dv = \cosh x\, dx$. Then $du = (n-1) \sinh x \cosh^{n-2} x\, dx$ and $v = \sinh x$, so

$$\begin{aligned} I_n &= \sinh x \cosh^{n-1} x - (n-1) \int \sinh^2 x \cosh^{n-2} x\, dx \\ &= \sinh x \cosh^{n-1} x - (n-1)I_n + (n-1)I_{n-2}. \end{aligned}$$

Thus $nI_n = \sinh x \cosh^{n-1} x + (n-1)I_{n-2}$, giving

$$I_n = \frac{\cosh^{n-1} x \sinh x}{n} + \frac{n-1}{n} I_{n-2}.$$

12b)

$$I_0 = x + c, \qquad I_2 = \frac{\sinh x \cosh x}{2} + \frac{x}{2} + c.$$

Then

$$\begin{aligned} I_4 &= \frac{\sinh x \cosh^3 x}{4} + \frac{3}{4} I_2 \\ &= \frac{\sinh x \cosh^3 x}{4} + \frac{3 \sinh x \cosh x}{8} + \frac{3x}{8} + c, \\ I_6 &= \frac{\sinh x \cosh^5 x}{6} + \frac{5}{6} I_4 \\ &= \frac{\sinh x \cosh^5 x}{6} + \frac{5 \sinh x \cosh^3 x}{24} + \frac{5 \sinh x \cosh x}{16} + \frac{5x}{16} + c. \end{aligned}$$

13a)

Let $I_n = \int x^n\sqrt{1+x}\,dx$. By integration by parts, take $u = x^n$ and $dv = \sqrt{1+x}\,dx$. Then $du = nx^{n-1}\,dx$ and $v = \frac{2}{3}(1+x)^{3/2}$. Hence

$$\begin{aligned} I_n &= \frac{2}{3}x^n(1+x)^{3/2} - \frac{2n}{3}\int x^{n-1}(1+x)^{3/2}\,dx \\ &= \frac{2}{3}x^n(1+x)^{3/2} - \frac{2n}{3}(I_n + I_{n-1}). \end{aligned}$$

Rearranging gives

$$I_n = \frac{2x^n(1+x)^{3/2}}{2n+3} - \frac{2n}{2n+3}I_{n-1},$$

as required.

13b)

$$I_0 = \int \sqrt{1+x}\,dx = \frac{2}{3}(1+x)^{3/2} + c.$$

Then

$$\begin{aligned} I_1 &= \frac{2x(1+x)^{3/2}}{5} - \frac{2}{5}I_0, \\ I_2 &= \frac{2x^2(1+x)^{3/2}}{7} - \frac{4}{7}I_1 \\ &= \frac{2x^2(1+x)^{3/2}}{7} - \frac{8x(1+x)^{3/2}}{35} + \frac{16(1+x)^{3/2}}{105} + c \\ &= \frac{(1+x)^{3/2}}{105}(30x^2 - 24x + 16) + c. \end{aligned}$$

14a)

$$\begin{aligned} I_n &= \int \tan^{n-2}x\tan^2 x\,dx \\ &= \int \tan^{n-2}x(\sec^2 x - 1)\,dx \\ &= \frac{\tan^{n-1}x}{n-1} - I_{n-2}, \end{aligned}$$

as required.

14b)

$$\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx = -\ln|\cos x| + c = \ln|\sec x| + c.$$

14c)

$$\begin{aligned} I_1 &= \ln|\sec x| + c, \\ I_3 &= \frac{1}{2}\tan^2 x - I_1, \end{aligned}$$

$$\begin{aligned} I_5 &= \frac{1}{4}\tan^4 x - I_3 \\ &= \frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x + \ln|\sec x| + c. \end{aligned}$$

15a)

Let $I_{m,n} = \int x^m (\ln x)^n \, dx$. By integration by parts, take $u = (\ln x)^n$ and $dv = x^m \, dx$. Then $du = \frac{n}{x}(\ln x)^{n-1} \, dx$ and $v = \frac{x^{m+1}}{m+1}$. Thus

$$\begin{aligned} I_{m,n} &= \frac{x^{m+1}(\ln x)^n}{m+1} - \frac{n}{m+1}\int x^m (\ln x)^{n-1} \, dx \\ &= \frac{x^{m+1}(\ln x)^n}{m+1} - \frac{n}{m+1} I_{m,n-1}, \end{aligned}$$

as required.

15b)

For $m = 3$, $n = 0$, $I_{3,0} = x^4/4 + c$. Hence

$$I_{3,1} = \frac{x^4}{4}\ln x - \frac{1}{4}I_{3,0} = \frac{x^4}{4}\ln x - \frac{x^4}{16} + c.$$

16a)

Let $I_n = \int_0^1 \frac{x^n}{\sqrt{1-x}} \, dx$. Integrate by parts with $u = x^n$ and $dv = (1-x)^{-1/2} \, dx$. Then $du = nx^{n-1} \, dx$ and $v = -2\sqrt{1-x}$. The boundary term is zero, so

$$\begin{aligned} I_n &= 2n\int_0^1 x^{n-1}\sqrt{1-x} \, dx \\ &= 2n\int_0^1 \frac{x^{n-1}(1-x)}{\sqrt{1-x}} \, dx \\ &= 2nI_{n-1} - 2nI_n. \end{aligned}$$

Therefore $(1+2n)I_n = 2nI_{n-1}$, and

$$I_n = \frac{2n}{1+2n} I_{n-1}.$$

16b)

$$I_0 = \int_0^1 \frac{1}{\sqrt{1-x}} \, dx = \left[-2\sqrt{1-x}\right]_0^1 = 2.$$

Then

$$\begin{aligned} I_1 &= \frac{2}{3}I_0 = \frac{4}{3}, \\ I_2 &= \frac{4}{5}I_1 = \frac{16}{15}, \\ I_3 &= \frac{6}{7}I_2 = \frac{32}{35}, \\ I_4 &= \frac{8}{9}I_3 = \frac{256}{315}. \end{aligned}$$

17a)

Let $I_n = \int_0^{\pi/2} x^n \sin x \, dx$. Integrating by parts with $u = x^n$ and $dv = \sin x \, dx$ gives

$$I_n = [-x^n \cos x]_0^{\pi/2} + n \int_0^{\pi/2} x^{n-1} \cos x \, dx = n \int_0^{\pi/2} x^{n-1} \cos x \, dx.$$

Integrating the remaining integral by parts,

$$\begin{aligned} \int_0^{\pi/2} x^{n-1} \cos x \, dx &= \left[x^{n-1} \sin x\right]_0^{\pi/2} - (n-1) \int_0^{\pi/2} x^{n-2} \sin x \, dx \\ &= \left(\frac{\pi}{2}\right)^{n-1} - (n-1) I_{n-2}. \end{aligned}$$

Thus

$$I_n = n \left(\frac{\pi}{2}\right)^{n-1} - n(n-1) I_{n-2}.$$

17b)

$$I_0 = \int_0^{\pi/2} \sin x \, dx = 1, \qquad I_1 = \int_0^{\pi/2} x \sin x \, dx = 1.$$

Hence $I_2 = \pi - 2$ and

$$\begin{aligned} I_4 &= 4 \left(\frac{\pi}{2}\right)^3 - 12 I_2 \\ &= \frac{\pi^3}{2} - 12\pi + 24 = \frac{1}{2}(48 - 24\pi + \pi^3). \end{aligned}$$

18a)

For $n \geq 2$, the reduction formula for cosine gives

$$I_n = \left[\frac{\sin x \cos^{n-1} x}{n}\right]_0^{2\pi} + \frac{n-1}{n} I_{n-2} = \frac{n-1}{n} I_{n-2}.$$

Since $I_1 = \int_0^{2\pi} \cos x \, dx = 0$, repeated application shows that $I_n = 0$ whenever n is odd.

18b)

For an even power, repeated application gives

$$\begin{aligned} I_{2n} &= \frac{2n-1}{2n} I_{2n-2} \\ &= \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} I_{2n-4} \\ &= \left(\frac{2n-1}{2n}\right)\left(\frac{2n-3}{2n-2}\right) \cdots \left(\frac{3}{4}\right)\left(\frac{1}{2}\right) I_0. \end{aligned}$$

As $I_0 = \int_0^{2\pi} 1 \, dx = 2\pi$, this gives the stated result.

19a)

For $n \geq 2$, the reduction formula for sine gives

$$I_n = \left[-\frac{\sin^{n-1} x \cos x}{n}\right]_0^{2\pi} + \frac{n-1}{n} I_{n-2} = \frac{n-1}{n} I_{n-2}.$$

Since $I_1 = \int_0^{2\pi} \sin x \, dx = 0$, repeated application shows that $I_n = 0$ whenever n is odd.

19b)

For an even power, repeated application gives

$$\begin{aligned} I_{2n} &= \frac{2n-1}{2n} I_{2n-2} \\ &= \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} I_{2n-4} \\ &= \left(\frac{2n-1}{2n}\right)\left(\frac{2n-3}{2n-2}\right) \cdots \left(\frac{3}{4}\right)\left(\frac{1}{2}\right) I_0. \end{aligned}$$

As $I_0 = \int_0^{2\pi} 1\,dx = 2\pi$, this gives the stated result.

20a)

Using $\sec^2 x = 1 + \tan^2 x$ and integration by parts,

$$\begin{aligned} I_n &= \int_0^{\pi/3} \sec^{n-2} x \sec^2 x\,dx \\ &= \Big[\sec^{n-2} x \tan x\Big]_0^{\pi/3} - (n-2)\int_0^{\pi/3} \sec^{n-2} x \tan^2 x\,dx \\ &= 2^{n-2}\sqrt{3} - (n-2)\int_0^{\pi/3} \sec^{n-2} x(\sec^2 x - 1)\,dx \\ &= 2^{n-2}\sqrt{3} - (n-2)(I_n - I_{n-2}). \end{aligned}$$

Therefore $(n-1)I_n = 2^{n-2}\sqrt{3} + (n-2)I_{n-2}$.

20b)

$$I_0 = \frac{\pi}{3}, \qquad I_2 = \sqrt{3}.$$

Applying the recurrence,

$$\begin{aligned} 3I_4 &= 4\sqrt{3} + 2I_2 = 6\sqrt{3}, \\ 5I_6 &= 16\sqrt{3} + 4I_4 = 24\sqrt{3}, \\ 7I_8 &= 64\sqrt{3} + 6I_6 = \frac{464\sqrt{3}}{5}. \end{aligned}$$

Thus $I_8 = \dfrac{464\sqrt{3}}{35}$.

21a)

Write $I_n = \int_0^{\pi/3} \sin^{n-1} x \sin x\,dx$. Integration by parts with $u = \sin^{n-1} x$ and $dv = \sin x\,dx$ gives

$$\begin{aligned} I_n &= \Big[-\sin^{n-1} x \cos x\Big]_0^{\pi/3} + (n-1)\int_0^{\pi/3} \sin^{n-2} x \cos^2 x\,dx \\ &= -\left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right)^{n-1} + (n-1)(I_{n-2} - I_n). \end{aligned}$$

Rearranging,

$$I_n = \frac{n-1}{n} I_{n-2} - \frac{1}{2n}\left(\frac{\sqrt{3}}{2}\right)^{n-1}.$$

21b)

$$I_0 = \frac{\pi}{3}, \qquad I_2 = \frac{\pi}{6} - \frac{\sqrt{3}}{8}.$$

Hence

$$I_4 = \frac{3}{4}I_2 - \frac{1}{8}\left(\frac{\sqrt{3}}{2}\right)^3 = \frac{\pi}{8} - \frac{9\sqrt{3}}{128},$$
$$I_6 = \frac{5}{6}I_4 - \frac{1}{12}\left(\frac{\sqrt{3}}{2}\right)^5 = \frac{5\pi}{48} - \frac{9\sqrt{3}}{64}.$$

21c)

Since $y = \sin^6 x$ is symmetrical about $x = 0$,

$$\int_{-\pi/3}^{\pi/3} \sin^6 x\, dx = 2I_6 = \frac{5\pi}{24} - \frac{9\sqrt{3}}{32}.$$

22)

Using symmetry and the substitution $x = 3t$,

$$\int_{-\pi/2}^{\pi/2} \cos^6(3t)\, dt = \frac{1}{3}\int_{-3\pi/2}^{3\pi/2} \cos^6 x\, dx.$$

The interval has length 3π, and $\cos^6 x$ has period π, so this is $\int_0^{\pi} \cos^6 x\, dx = 2I_6$. From the recurrence, $I_0 = \pi/2$, $I_2 = \pi/4$, $I_4 = 3\pi/16$, and $I_6 = 5\pi/32$. Therefore the value is $5\pi/16$.

23a)

Let $I_n = \int_0^1 (1-x^2)^n\, dx$. Integration by parts with $u = (1-x^2)^n$ and $dv = dx$ gives

$$\begin{aligned} I_n &= \Big[x(1-x^2)^n\Big]_0^1 + 2n\int_0^1 x^2(1-x^2)^{n-1}\, dx \\ &= 2n\int_0^1 \left((1-x^2)^{n-1} - (1-x^2)^n\right) dx \\ &= 2n(I_{n-1} - I_n). \end{aligned}$$

Thus $I_n = \dfrac{2n}{2n+1} I_{n-1}$.

23b)

$$I_0 = \int_0^1 1\, dx = 1.$$

Then

$$I_6 = \frac{12}{13}\cdot\frac{10}{11}\cdot\frac{8}{9}\cdot\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3} I_0 = \frac{1024}{3003}.$$

Since $(1-x^2)^6$ is symmetrical about $x = 0$,

$$\int_{-1}^{1} (1-x^2)^6\, dx = 2I_6 = \frac{2048}{3003}.$$

24a)

Let $I_n = \int_0^{\pi/2} \cos^{2n} x \sin^2 x \, dx$ and $J_n = \int_0^{\pi/2} \cos^n x \, dx$. Integration by parts gives

$$\begin{aligned} I_n &= \left[-\frac{x \cos^{2n+1} x}{2n+1} \right]_0^{\pi/2} + \frac{1}{2n+1} \int_0^{\pi/2} \cos^{2n+2} x \, dx \\ &= \frac{1}{2n+1} \int_0^{\pi/2} \cos^{2n} x (1 - \sin^2 x) \, dx \\ &= \frac{1}{2n+1} J_{2n} - \frac{1}{2n+1} I_n. \end{aligned}$$

Consequently $I_n = \dfrac{1}{2n+2} J_{2n}$. Also $J_{2n} = \dfrac{2n-1}{2n} J_{2n-2}$, while $I_{n-1} = \dfrac{1}{2n} J_{2n-2}$. Therefore

$$I_n = \frac{2n-1}{2n+2} I_{n-1}.$$

24b)

Starting with $I_0 = \int_0^{\pi/2} \sin^2 x \, dx = \pi/4$,

$$\begin{aligned} I_6 &= \frac{11}{14} I_5 = \frac{11}{14} \cdot \frac{9}{12} I_4 = \frac{33}{56} I_4 = \frac{33}{56} \cdot \frac{7}{10} I_3 \\ &= \frac{33}{80} I_3 = \frac{33}{80} \cdot \frac{5}{8} I_2 = \frac{33}{128} I_2 = \frac{33}{128} \cdot \frac{3}{6} I_1 \\ &= \frac{33}{256} I_1 = \frac{33}{256} \cdot \frac{1}{4} I_0 = \frac{33\pi}{4096}. \end{aligned}$$

24c)

Since $y = \cos^{12} x \sin^2 x$ is symmetrical about $x = 0$,

$$\int_{-\pi/2}^{\pi/2} \cos^{12} x \sin^2 x \, dx = 2I_6 = \frac{33\pi}{2048}.$$

25a)

Using $\tanh^2 x = 1 - \operatorname{sech}^2 x$ and integration by parts,

$$\begin{aligned} I_n &= \int_0^{\ln 3} \tanh^{n-2} x (1 - \operatorname{sech}^2 x) \, dx \\ &= I_{n-2} - \int_0^{\ln 3} \tanh^{n-2} x \operatorname{sech}^2 x \, dx. \end{aligned}$$

For the remaining integral, take $u = \tanh^{n-2} x$ and $dv = \operatorname{sech}^2 x \, dx$. Then

$$\begin{aligned} \int_0^{\ln 3} \tanh^{n-2} x \operatorname{sech}^2 x \, dx &= \left[\tanh^{n-1} x \right]_0^{\ln 3} - (n-2) \int_0^{\ln 3} \tanh^{n-2} x \operatorname{sech}^2 x \, dx \\ &= \left(\frac{4}{5} \right)^{n-1} - (n-2) \int_0^{\ln 3} (1 - \tanh^2 x) \tanh^{n-2} x \, dx \\ &= \left(\frac{4}{5} \right)^{n-1} - (n-2)(I_{n-2} - I_n). \end{aligned}$$

Substitution and rearrangement give $I_n = I_{n-2} - \dfrac{1}{n-1} \left(\dfrac{4}{5} \right)^{n-1}$.

25b)

$$I_0 = \ln 3, \qquad I_2 = I_0 - \frac{4}{5} = \ln 3 - \frac{4}{5}.$$

Therefore

$$I_4 = I_2 - \frac{1}{3}\left(\frac{4}{5}\right)^3 = \ln 3 - \frac{4}{5} - \frac{64}{375} = \ln 3 - \frac{364}{375}.$$

26a)

Integration by parts with $u = x^n$ and $dv = e^{ax}\,dx$ gives

$$I_n = \left[\frac{x^n e^{ax}}{a}\right]_0^1 - \frac{n}{a}\int_0^1 x^{n-1}e^{ax}\,dx = \frac{e^a}{a} - \frac{n}{a}I_{n-1}.$$

26b)

Here $a = \frac{1}{4}$. First,

$$I_0 = \int_0^1 e^{x/4}\,dx = 4e^{1/4} - 4.$$

Then $I_1 = 4e^{1/4} - 4I_0$, and

$$\begin{aligned} I_2 &= 4e^{1/4} - 8I_1 \\ &= 4e^{1/4} - 8(4e^{1/4} - 4I_0) \\ &= -28e^{1/4} + 32I_0 \\ &= 100e^{1/4} - 128. \end{aligned}$$

27a)

Integration by parts with $u = (a - x^2)^n$ and $dv = dx$ gives

$$\begin{aligned} I_n &= \left[x(a-x^2)^n\right]_0^{\sqrt{a}} + 2n\int_0^{\sqrt{a}} x^2(a-x^2)^{n-1}\,dx \\ &= 2n\int_0^{\sqrt{a}} (a-(a-x^2))(a-x^2)^{n-1}\,dx \\ &= 2naI_{n-1} - 2nI_n. \end{aligned}$$

Hence $I_n = \dfrac{2na}{1+2n}I_{n-1}$.

27b)

$$I_0 = \int_0^4 1\,dx = 4, \qquad I_n = \frac{32n}{1+2n}I_{n-1}.$$

Thus

$$I_3 = \frac{96}{7}I_2 = \frac{96}{7}\cdot\frac{64}{5}I_1 = \frac{6144}{35}I_1 = \frac{6144}{35}\cdot\frac{32}{3}I_0 = \frac{262144}{35}.$$

28a)

Set $A = (n-1)x$ and $B = x$. Then $A + B = nx$ and $A - B = (n-2)x$. The identity gives

$$\cos(nx) + \cos((n-2)x) = 2\cos((n-1)x)\cos x,$$

so, dividing by $\cos x$,

$$\frac{\cos nx}{\cos x} = 2\cos((n-1)x) - \frac{\cos((n-2)x)}{\cos x}.$$

28b)

Integrating the identity from 0 to π,

$$\begin{aligned} I_n &= 2\int_0^\pi \cos((n-1)x)\,dx - I_{n-2} \\ &= \left[\frac{2}{n-1}\sin((n-1)x)\right]_0^\pi - I_{n-2} = -I_{n-2}. \end{aligned}$$

28c)

$$I_1 = \int_0^\pi 1\,dx = \pi, \qquad I_7 = -I_5 = I_3 = -I_1 = -\pi.$$

28d)

There are $\frac{53-1}{2} = 26$ applications of the recurrence, so

$$I_{53} = (-1)^{26} I_1 = \pi.$$

29a)

Take $u = x^n$ and $dv = (a-x)^m\,dx$, so $du = nx^{n-1}\,dx$ and $v = -\dfrac{(a-x)^{m+1}}{m+1}$. Then

$$\begin{aligned} I_{m,n} &= \left[-\frac{x^n(a-x)^{m+1}}{m+1}\right]_0^k + \frac{n}{m+1}\int_0^k x^{n-1}(a-x)^{m+1}\,dx \\ &= -\frac{k^n(a-k)^{m+1}}{m+1} + \frac{n}{m+1} I_{m+1,n-1}. \end{aligned}$$

29b)

Using the recurrence with $k = 1$, $a = 2$, $m = -3$ and $n = 4$,

$$\begin{aligned} I_{-3,4} &= \frac{1}{2} - 2I_{-2,3}, \\ I_{-2,3} &= 1 - 3I_{-1,2}, \\ I_{-3,4} &= -\frac{3}{2} + 6I_{-1,2}. \end{aligned}$$

The integral for $I_{-1,2}$ is evaluated separately:

$$I_{-1,2} = \int_0^1 \frac{x^2}{2-x}\,dx = \int_0^1 \left(-x - 2 + \frac{4}{2-x}\right)\,dx$$

$$= \left[-\frac{x^2}{2} - 2x - 4\ln|2-x| \right]_0^1$$
$$= -\frac{5}{2} + 4\ln 2.$$

Therefore

$$I_{-3,4} = -\frac{3}{2} + 6\left(-\frac{5}{2} + 4\ln 2\right) = 24\ln 2 - \frac{33}{2}.$$

30a)

$$\lim_{x\to 0^+} x\ln x = \lim_{x\to 0^+} \frac{\ln x}{1/x} = \lim_{x\to 0^+} \frac{1/x}{-1/x^2} = \lim_{x\to 0^+} (-x) = 0.$$

30b)

Integration by parts on $[a, 1]$ gives

$$\int_a^1 (\ln x)^n \, dx = [x(\ln x)^n]_a^1 - n\int_a^1 (\ln x)^{n-1} \, dx.$$

For $n \geq 1$, $a(\ln a)^n = (a\ln a)(\ln a)^{n-1} \to 0$ as $a \to 0^+$; the endpoint at 1 is also zero. Taking the limit yields $I_n = -nI_{n-1}$.

30c)

For $n = 0$, $I_0 = \int_0^1 1 \, dx = 1 = (-1)^0 0!$. Suppose $I_k = (-1)^k k!$. Then

$$I_{k+1} = -(k+1)I_k = -(k+1)(-1)^k k! = (-1)^{k+1}(k+1)!.$$

Thus $I_n = (-1)^n n!$ for all integers $n \geq 0$.

30d)

$$\int_0^1 (\ln x)^4 \, dx = I_4 = (-1)^4 4! = 24.$$